

Coding & Solution

Date	30 June 2026
Team ID	xxxxxx
Project Name	Rising Waters
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/mamidalaswathi2006-alt/Rising_Waters_Project
Programming Language(s)	Python, JavaScript, HTML5, CSS3
Framework(s) Used	Flask, Scikit-learn (Machine Learning Framework)

Key Features Implemented	• Machine learning-based flood prediction using historical weather data. • Implemented and compared Decision Tree, Random Forest, KNN, and XGBoost algorithms. • Predicts flood risk using features such as annual rainfall, cloud visibility, and seasonal rainfall. • Best-performing XGBoost model selected with 96.55% prediction accuracy. • Integrated the trained model into a Flask web application for easy flood-risk prediction. • Simple user interface for entering weather data and viewing prediction results. • Supports early flood warning for evacuation planning and disaster response. • Designed for deployment on IBM Cloud for scalable and remote access.
Pending / Incomplete Features	
Setup / Run Instructions	1. Clone the GitHub Repository 2. Open the project folder in Visual Studio Code 3. Install the required dependencies using: pip install -r requirements.txt 4. Run the application using: cd Flask, & python app.py 5. Open the application in your browser at: 6. Or access the deployed application at: Swathi7892.pythonanywhere.com

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes

2	Meaningful variable and function names are used	Yes
---	---	-----

Code Quality Checklist

3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The "Rising Waters" system has been completely configured and locally tested with standard simulated IoT telemetry benchmarks. The Python ML predictive models show 93% accuracy in estimating flood windows. Code layout rigidly adheres to production standards, and all core components are modularly optimized for smooth deployment transitions.